**Unit no. º 3**

# LINES IN THE PLANE

1-16 October 2026

## I Direction vector and Cartesian equation

**Definition**

A direction vector of a line is a vector **non-zero** with the direction of this line. If $A \neq B$, $AB$ directs $(AB)$. Every line has an equation $ax + by + c = 0$ with $(a, b) \neq (0, 0)$. A direction vector is $\begin{pmatrix} -b \\ a \end{pmatrix}$.

**Example 3.1**

For $A(-2; 4)$ and $B(6; 2)$, calculate $\overrightarrow{AB}$ then verify that $\vec{u}(4; -1)$ directs $(AB)$.

**Solution:**

$\overrightarrow{AB} = (8; -2) = 2\vec{u}$. The vector $\vec{u}$ is non-zero and collinear with $\overrightarrow{AB}$; it therefore directs $(AB)$.

**Establish an equation using the determinant**

The line through $A(x_A; y_A)$ and directed by $\vec{u}(\alpha; \beta) \neq 0$ has equation

$$\beta(x - x_A) - \alpha(y - y_A) = 0.$$

**Write a proof.**

**Solution:**

A point $M(x; y)$ lies on the line if and only if $\overrightarrow{AM}$ is collinear with $\vec{u}$, including for $M = A$. Now

$$\det(\overrightarrow{AM}, \vec{u}) = (x - x_A)\beta - (y - y_A)\alpha.$$

Setting the determinant to zero gives the stated equation. Expanding, $a = \beta,\ b = -\alpha$ and $c = -\beta x_A + \alpha y_A$; since $\vec{u} \neq \vec{0}$, $(a, b) \neq (0, 0)$.

**Watch out.** The zero vector does not direct any line. The sign of the determinant depends on the order of the vectors, but whether it is zero does not depend on them.

## II Determine an equation of a line

**Method**

From a point and a direction vector: choose $a, b$ then replace $x, y$ by the coordinates of the point to find $c$. From two distinct points: first calculate a direction vector.

**Example 3.2**

Determine a Cartesian equation of the line passing through $A(2; -1)$ and directed by $\vec{u}(3; 2)$.

**Solution:**

$a = 2,\ b = -3$ work. The line therefore has an equation 2 $x - 3y + c$ = 0. Since $A$ belongs to it, $4 + 3 + c$ = 0, hence $c = -7$. An equation is 2 $x - 3y - 7$ = 0. Check: 2 $\times 2 - 3 \times (-1) - 7 = 0$ and (3; 2) is indeed a direction vector.

**Example 3.3**

Determine a Cartesian equation of ( $AB$) for $A$(3; 3) and $B(2; -2)$.

**Solution:**

$\overrightarrow{AB} = (-1; -5)$ directs the line. We can choose ( $a, b) = (5, -1)$. With $A,\ 15 - 3 + c$ = 0, therefore $c = -12$. Thus 5 $x - y - 12$ = 0. For $B,\ 10 + 2 - 12$ = 0, which confirms the result.

**Example 3.4**

Is point $P$(1; 2) lie on $d : 2x - 3y$ + 4 = 0? Does the point $Q$(2; 1) lie on it?

**Solution:**

For $P,\ 2 - 6$ + 4 = 0: yes. For $Q,\ 4 - 3 + 4 = 5 \neq 0$: no.

## III Slope-intercept equation, slope and affine function

**Definitions**

A non-vertical line has a unique equation of the form $y = mx + p$. $m$ is the slope (gradient ), $p$ the y-intercept. It represents the affine function $x \mapsto mx + p$ ; $(1; m)$ is a direction vector. A vertical line has an equation $x = k$ ; it is not the graph of a function of $x$. A horizontal line has an equation $y = k$.

**Slope calculation**

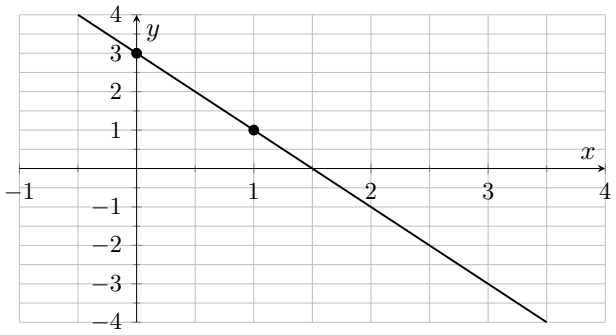
If $x_A \neq x_B$, the slope of ( $AB$) is $m = \dfrac{y_B - y_A}{x_B - x_A}$. To find $p$, then we use $y_A = mx_A + p$.

**Example 3.5**

Determine the equation y = mx + p of 2 $x + 3y - 1 = 0$, then draw $d : y = -2x + 3$.

**Solution:**

$3y = -2x + 1$, therefore $y = -\frac{2}{3}x + \frac{1}{3}$. For $d$, plot (0; 3) and (1; 1), then draw the line; a displacement of 1 along the x-axis corresponds to $-2$ on the y-axis.




**Example 3.6**

Find the affine function $f$ such that $f(0) = 3$ and $f(-1) = 5$, then the affine function $g$ whose graph passes via $C(-1; 2)$ and $E(5; -2)$.

**Solution:**

$f(x) = -2x + 3$, since $m = (3 - 5)/(0 + 1) = -2$. For $g$, $m = (-2 - 2)/(5 + 1) = -2/3$ and $p = 2 - 2/3 = 4/3$, so $g(x) = -\frac{2}{3}x + \frac{4}{3}$.

## IV Relative positions of two lines

**Properties**

Two lines are intersecting, strictly parallel or coincident. For $y = mx + p$ and $y = m'x + p'$ :

· if $m \neq m'$, they intersect;

· if $m = m'$ & $p \neq p'$, they are strictly parallel;

· if $m = m'$ & $p = p'$, they coincide.

For Cartesian equations, we compare directions using the determinant. If the directions are collinear, we then test whether a point belongs to both to distinguish coincident lines from strictly parallel lines.

**Example 3.7**

Determine an equation of $d$ through $A(2; 3)$ and parallel to $d' : x - 3y - 3 = 0$.

**Solution:**

$d'$ has direction vector (3; 1). We look for $d : x - 3y + c = 0$. Since $A \in d$, $2 - 9 + c = 0$, so $c = 7$. Thus $d : x - 3y + 7 = 0$. The two lines are strictly parallel, because the constants differ for the same coefficients of $x, y$.

**Example 3.8**

Study the relative positions:

1. $2x - y + 1 = 0$ and $4x - 2y + 2 = 0$ ;
2. $x = 2$ and $x = -1$ ;
3. $y = 2x - 3$ and $y = -x + 3$.

**Solution:**

1. The second equation is twice the first: the lines coincide. 2. Two distinct vertical lines: strictly parallel. 3. The slopes 2 and $-1$ differ: the lines intersect. Their intersection satisfies $2x - 3 = -x + 3$, so $x = 2$ and $y = 1$.

## V Intersection: solving by substitution

**Definition**

Solving a system of two equations in $x, y$, means finding all pairs $(x; y)$ satisfying the two equations simultaneously. For two equations of lines: an intersection gives a solution pair; strictly parallel lines give no solution; coincident lines give infinitely many solutions.

**Substitution**

Make one unknown the subject of one equation, substitute its expression into the other, solve, then find the second unknown. Check the pair in both original equations.

**Example 3.9**

Solve $\begin{cases} -2x + y + 3 = 0, \\ 3x - 5y - 2 = 0. \end{cases}$

**Solution:**

The first equation gives $y = 2x - 3$. Substituting into the second: $3x - 5(2x - 3) - 2 = 0$, i.e. $-7x + 13 = 0$. Thus $x = 13/7$ then $y = 5/7$. The solution set is $S = \{(13/7; 5/7)\}$. The substitutions give $(-26 + 5 + 21)/7 = 0$ and $(39 - 25 - 14)/7 = 0$.

**Example 3.10**

Solve by substitution:

$$\text{(a)} \begin{cases} -2x + y + 3 = 0 \\ x - 4y + 5 = 0 \end{cases} \qquad \text{(b)} \begin{cases} 2x + 3y - 10 = 0 \\ 4x + 5y - 16 = 0. \end{cases}$$

**Solution:**

(a) $y = 2x - 3$ then $x - 4(2x - 3) + 5 = 0$, hence $x = 17/7$, $y = 13/7$.
(b) $x = (10 - 3y)/2$ ; $2(10 - 3y) + 5y - 16 = 0$, therefore $y = 4$, then $x = -1$.

## VI Linear combination and Python

**Linear combination**

Multiply the equations by non-zero numbers to obtain opposite coefficients of one unknown; add them to eliminate this unknown. Keep one equation of the system to calculate the other unknown.

**Example 3.11**

Solve $\begin{cases} 2x+3y+1=0 \\ 3x-4y-2=0. \end{cases}$ Then redo the two systems from the previous example by elimination.

**Solution:**

Multiply the first equation by 3 and the second by $-2$ gives 6 $x+9y$ +3 = 0 and $-6x+8y+4=0$. The sum gives 17 $y$ + 7 = 0, hence $y=-7/17$. Then $2x-21/17$ + 1 = 0, therefore $x=2/17$. For (a), add four times the first equation to the second: $-7x$ + 17 = 0, therefore $(x;y)=(17/7;13/7)$. For (b), subtract twice the first from the second: $-y$ + 4 = 0, therefore $(x;y)=(-1;4)$.

```
def determinant(u, v):
    return u[0]*v[1] - u[1]*v[0]

def alignes(A, B, C):
    AB = (B[0]-A[0], B[1]-A[1])
    AC = (C[0]-A[0], C[1]-A[1])
    return determinant(AB, AC) == 0
```

**Example 3.12**

Predict `alignes((0,0), (2,1), (6,3))` then `alignes((0,0), (2,1), (6,4))`.

**Solution:**

`True`, then `False` : the determinants are 0 and 2. The exact test suits integer coordinates here; floating-point approximations require taking numerical errors into account.